Research Assignment 2: Exercise 1
SQL Fundamentals SELECT & Filtering

① Selecting & Sorting
TABLE employees

Id	First_name	Last-name	department	salary	hire-date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	zulu	Finance	65000	2018-07-18	Durban
4	David	mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

1. Retrieve all columns from the employees table.

```
SELECT *
FROM employees;
```

Id	First-name	Last-name	department	Salary	hire-date	City
1	Alice	Green	IT	70000	2020-01-10	Johanne
2	Brian	Lee	HR	45000	2019-03-22	Capetown
3	Cathy	zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesb

2. Find all unique departments

```
SELECT DISTINCT Departments
FROM employees;
```

departments
IT
HR
Finance
marketing

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3. Retrieve first and last names ordered by salary descending.

```
SELECT first_name,
       last_name
FROM employees
ORDER By Salary DESC;
```

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. Retrieve the top 3 highest-paid employees.

```
SELECT id,
       first_name,
       last_name,
       Salary
FROM employees
ORDER By Salary DESC
LIMIT 3;
```

Id	First name	last_name	Salary
5	Eva	Naidoo	70000
1	Alice	Green	70000
3	Cathy	Zulu	65000

2 Filtering with WHERE

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5 Find employees in the IT department.

```
SELECT id,
        first_name,
        last_name,
        department
FROM employees
WHERE department = 'IT';
```

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

7. Find employees in HR or marketing

```
SELECT id,
        first_name,
        last_name,
        department
FROM employees
WHERE department = 'HR' or department = 'Marketing';
```

id	last_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

6 Find employees in finance with salary > 60000

```
SELECT id,
        first_name,
        last_name,
        department,
        salary
```

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```
FROM employees
WHERE department = 'Finance' AND salary > 60000
```

Id	First-name	last-name	Salary
3	Cathy	Finance	65000

8. Find employees not in IT

```
SELECT Id,
       first-name,
       last-name,
       department
FROM employees
WHERE department != 'IT';
```

Id	First-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. Find employees in IT, HR, or Finance using IN

```
SELECT Id,
       first-name,
       last-name,
       department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

Id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

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10. Find employees in IT with salary >65000 and City johannesburg

```
SELECT Id,
       First_name,
       last_name,
       department,
       Salary,
       City
FROM employees
WHERE department = 'IT'
AND Salary > 65000
AND City = 'Johannesburg';
```

Id	first_name	last_name	department	Salary	City
1	Alice	Green	IT	70000	johannesburg
5	Eva	Naidoo	IT	72000	johannesburg

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